

ELEC4631 Solutions Tutorial 7

1. (a) We want the observer eigenvalues to be $-4 \pm 3j$. So the desired characteristic polynomial is $R(s) = (s+4-3j)(s+4+3j) = (s+4)^2 + 9 = s^2 + 8s + 25$. Then we can compute $R\left(\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\right) = \begin{bmatrix} 34 & 0 \\ 0 & 18 \end{bmatrix}$. The observability matrix for the system is $\mathcal{O} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. So, using Ackermann's formula applied to the controllable pair (A^T, C^T) we have:

$$\begin{aligned} L &= R(A)\mathcal{O}^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \\ &= \begin{bmatrix} 17 \\ -9 \end{bmatrix}. \end{aligned}$$

(Indeed, you can check that the eigenvalues of $A - LC$ are $-4 \pm 3j$).

The observer equations are:

$$\begin{aligned} \dot{\xi}^o(t) &= \begin{bmatrix} -16 & -17 \\ 9 & 8 \end{bmatrix} \xi^o(t) + \begin{bmatrix} -16 \\ 10 \end{bmatrix} u(t) + \begin{bmatrix} 17 \\ -9 \end{bmatrix} y(t), \\ y^o(t) &= \xi^o(t). \end{aligned}$$

- (b) The block diagram is shown on the next page.

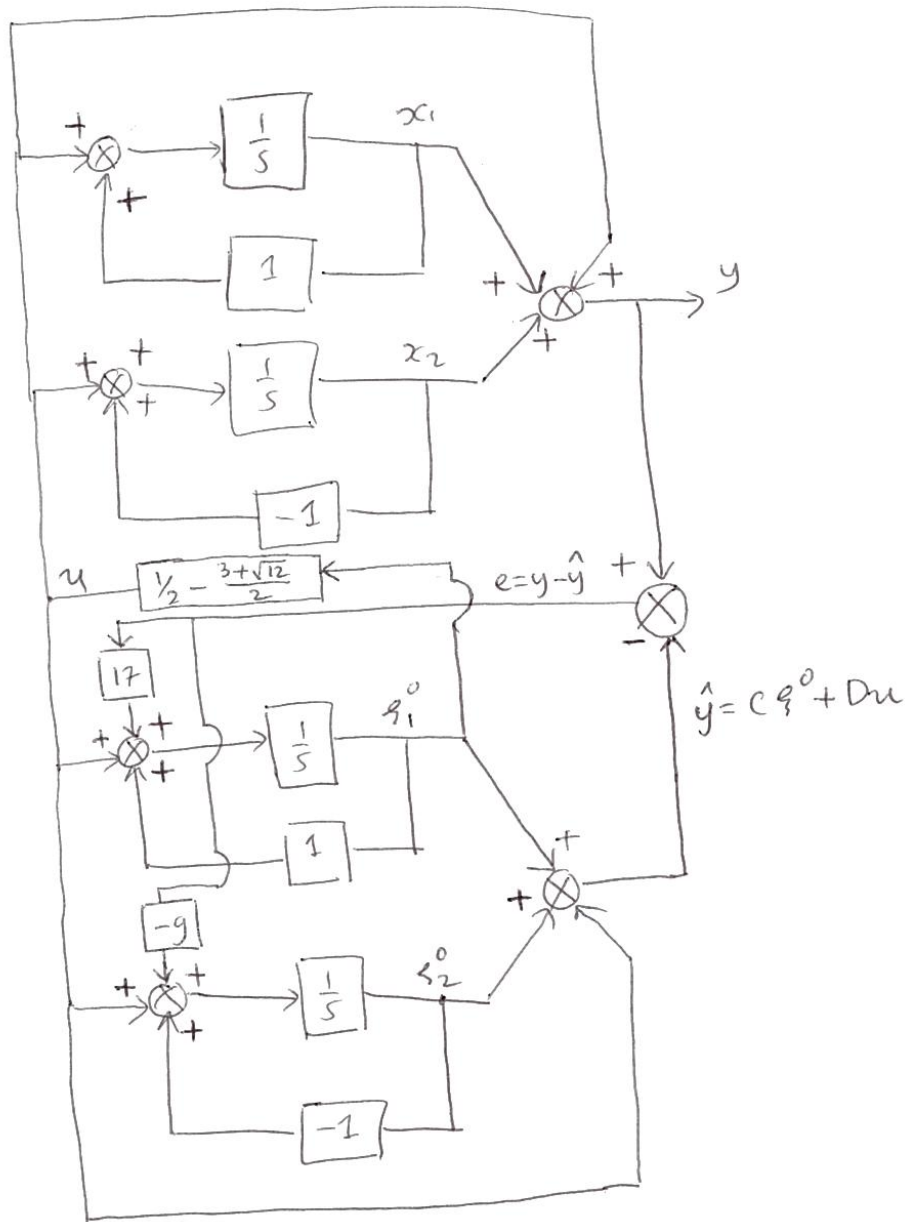


Figure 1: Sketch for Q1 (b).

2. The vector field for the system is

$$f(x) = \begin{bmatrix} -x_1 - x_2 \\ -x_1 - x_2^3 \end{bmatrix}.$$

An equilibrium point satisfies $f(x^*) = 0$, giving the two equations:

$$-x_1^* - x_2^* = 0, \quad (1)$$

$$-x_1^* - (x_2^*)^3 = 0. \quad (2)$$

Equation (1) gives $x_2^* = -x_1^*$. Substituting (1) into (2) gives $-x_1^* + (x_1^*)^3 = 0$. Solving this equation gives $x_1^* = 0$ or $x_1^* = \pm 1$. For $x_1^* = 0$ we obtain from (1) that $x_2^* = 0$, while for $x_1^* = \pm 1$ we obtain $x_2^* = \mp 1$. So, there are three equilibria: $(0, 0)$, $(-1, 1)$ and $(1, -1)$. The Jacobian at an equilibrium point x^* is:

$$\text{Jac}_{x^*}(f) = \begin{bmatrix} -1 & -1 \\ -1 & -3(x_2^*)^2 \end{bmatrix}.$$

We now check the stability of each equilibria.

For the equilibrium point, $x^* = (0, 0)$,

$$A = \text{Jac}_{x^*}(f) = \begin{bmatrix} -1 & -1 \\ -1 & 0 \end{bmatrix}.$$

We have $c_A(\lambda) = \lambda(\lambda + 1) - 1 = (\lambda + 1/2)^2 - 5/4$. The eigenvalues are $\lambda_1 = (-1 + \sqrt{5})/2$ and $\lambda_2 = (-1 - \sqrt{5})/2$. Since λ_1 is in RHP, by the indirect method of Lyapunov we know that $(0, 0)$ is an unstable equilibrium point.

We now note that the Jacobian matrix for the equilibria $x^* = (1, -1)$ and $x^* = (-1, 1)$ are the same, so both equilibria will have the same stability property. We have

$$A = \text{Jac}_{x^*}(f) = \begin{bmatrix} -1 & -1 \\ -1 & -3 \end{bmatrix}.$$

The characteristic equation is $c_A(\lambda) = (\lambda + 1)(\lambda + 3) - 1 = (\lambda + 2)^2 - 2$. The eigenvalues are $\lambda_1 = -2 + \sqrt{2}$ and $\lambda_2 = -2 - \sqrt{2}$. Since both eigenvalues are in the LHP, by the indirect method of Lyapunov we know that $x^* = (1, -1)$ and $x^* = (-1, 1)$ are asymptotically stable equilibria.

3. The state for the damped pendulum from Lecture 3 is (θ, ω) . The vector field is,

$$f(\theta, \omega) = \begin{bmatrix} \omega \\ -(g/r) \sin(\theta) - D\omega \end{bmatrix},$$

with g, r, D being positive parameters. We know that $(0, 0)$ and $(\pi, 0)$ are equilibria of the pendulum corresponding to the pendulum in the downward and inverted position, respectively. The Jacobian at an equilibrium point (θ^*, ω^*) is

$$\text{Jac}_{x^*}(f) = \begin{bmatrix} 0 & 1 \\ -(g/r) \cos(\theta^*) & -D \end{bmatrix}.$$

For the equilibrium point $(0, 0)$, we have that

$$A = \text{Jac}_{(0,0)}(f) = \begin{bmatrix} 0 & 1 \\ -g/r & -D \end{bmatrix},$$

with $c_A(\lambda) = \lambda(\lambda + D) + g/r = (\lambda + D/2)^2 + g/r - D^2/4$. The eigenvalues are $\lambda_1 = -D/2 + \sqrt{D^2/4 - g/r}$ and $\lambda_2 = -D/2 - \sqrt{D^2/4 - g/r}$. If $g/r < D^2/4$ then both eigenvalues are real and in the LHP. If $g/r = D^2/4$ there is only one eigenvalue which is repeated ($\lambda_1 = \lambda_2$) but it will be in the LHP. Finally, when $g/r > D^2/4$ there will be a pair of complex conjugate eigenvalues with real part equal to $-D/2$, so they lie in the LHP. So, for whatever values of the positive parameters g, r and D the eigenvalues are always in the LHP. Thus, by the indirect method of Lyapunov, $(0, 0)$ is an asymptotically stable equilibrium point.

For the equilibrium point $(\pi, 0)$, we have that

$$A = \text{Jac}_{(\pi,0)}(f) = \begin{bmatrix} 0 & 1 \\ g/r & -D \end{bmatrix},$$

with $c_A(\lambda) = \lambda(\lambda + D) - g/r = (\lambda + D/2)^2 - g/r - D^2/4$. The eigenvalues are $\lambda_1 = -D/2 + \sqrt{g/r + D^2/4}$ and $\lambda_2 = -D/2 - \sqrt{g/r + D^2/4}$. Since $\sqrt{g/r + D^2/4} > D/2$, λ_1 is in the RHP. So, for whatever values of the parameters g, r and D this eigenvalue is always in the RHP. Thus, by the indirect method of Lyapunov, $(\pi, 0)$ is an unstable equilibrium point.

4. (a) Setting $f(x^*) = 0$ and solving for x^* , we find that there is only one equilibrium point $x^* = (x_1^*, x_2^*, x_3^*) = (c/2, 1, -1)$.
- (b) Let $A = \text{Jac}_{x^*}(f)$. Expanding A based on expansion by minors of its last row gives $c_A(s) = \det(sI - A) = (s - x_1^* + c)(s(s + c/2) + 1) + x_3^*(s + c/2)$. Noting that $s - x_1^* + c = s + c/2$ (since $x_1^* = c/2$) we can rewrite this as $c_A(s) = (s + c/2)(s(s + c/2) + 1 + x_3^*)$.
- (c) For $c = 1$, $x^* = (1/2, 1, -1)$ and $c_A(s) = (s + 1/2)^2 s$. So, eigenvalues of A are $-1/2$ (with algebraic multiplicity 2) and 0. There is one non-repeated eigenvalue on imaginary axis, so A is marginally stable and the linearised system

about the point x^* is internally stable, but not internally asymptotically stable. Therefore, **no** conclusion can be made with the first/indirect method of Lyapunov regarding the stability of the equilibrium point x^* .

5. (a) The closed-loop transfer function is $H_{cl}(s) = \frac{C(s)P(s)}{1+C(s)P(s)} = \frac{\frac{1}{s}}{1+\frac{1}{s}} = \frac{1}{s+1}$.
- (b) In open-loop, set $e(t) = r$ and $u(t) = x_2(t)$, so we get that $\dot{x}_1(t) = u(t) = x_2(t)$ and $\dot{x}_2(t) = x_2(t) + r$ and the output is $y(t) = -x_1(t) + u(t) = -x_1(t) + x_2(t)$. Putting them together yields the state-space equation for the cascade of C and P

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} e(t),$$

$$y(t) = \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}.$$

The controllability matrix is $\mathcal{C} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$ which is invertible, so the cascade is controllable. The observability matrix is $\mathcal{O} = \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix}$ which is not invertible, so the cascade is not observable.

- (c) The closed-loop state-space equation can be obtained from the open loop one simply by substituting $e(t) = r - y(t) = r + x_1(t) - x_2(t)$. This gives:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} r,$$

$$y(t) = \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}.$$

The controllability matrix is $\mathcal{C} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ which is invertible, so the closed-loop system is controllable. The observability matrix is $\mathcal{O} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$ which is not invertible, so the closed-loop is not observable.

(d) The solution of the state-space equation is

$$\begin{aligned}
\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} &= e^{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} t} x_0 + r \int_0^t e^{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} (t-s)} \begin{bmatrix} 0 \\ 1 \end{bmatrix} ds, \\
&= -\frac{1}{2} \begin{bmatrix} -e^t - e^{-t} & -e^t + e^{-t} \\ -e^t + e^{-t} & -e^t - e^{-t} \end{bmatrix} x_0 + r \int_0^t \frac{1}{2} \begin{bmatrix} e^{t-s} - e^{-(t-s)} \\ e^{t-s} + e^{-(t-s)} \end{bmatrix} ds \\
y(t) &= \begin{bmatrix} -1 & 1 \end{bmatrix} e^{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} t} x_0 + r \int_0^t \begin{bmatrix} -1 & 1 \end{bmatrix} e^{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} (t-s)} \begin{bmatrix} 0 \\ 1 \end{bmatrix} ds, \\
&= \begin{bmatrix} -e^{-t} & e^{-t} \end{bmatrix} x_0 + r(1 - e^{-t}).
\end{aligned}$$

From the solution of the state-space equation it can be seen that $y(t) \rightarrow r$ as $t \rightarrow \infty$ but $|x_1(t)| \rightarrow \infty$ and $|x_2(t)| \rightarrow \infty$ as $t \rightarrow \infty$ for any x_0 (the term under the integral for $x(t)$ diverges for any x_0). So, the system can track the set-point r , but need not be internally stable. In the practice of control systems, care has to be taken to ensure this does not happen.